

Predicting the Acres Burned by Californian Wildfires Using a Small Dataset

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Abstract

The problem that we are attempting to solve is wildfires being very unpredictable in California. We are trying to alleviate this by predicting how much damage a given wildfire will cause, in terms of acres of land burned, given certain data points about the fire. Wildfires are becoming prevalent as global warming continues, and we must ramp up our efficiency to put them out, until the root problem is solved. This is a critical issue as wildfires cause severe amounts of economic, environmental, and human damage [1]. To begin, we took a data set of previous wildfires in California, and used circumstantial information such as latitude, longitude, whether this was a major incident, and more, to predict how many acres the fire will burn. We used this data in combination with an XGBoost model to get our final estimate. We ended up with a mean absolute error of about 3,300 acres burned. While this is not the best margin of error attainable, considering the average wildfire seems to hover around 10,000 acres, it is still significant given the restricted amount of data available to this study (1,637 rows). One major challenge indeed, as climate change induced wildfires are still relatively new phenomena, was for us to figure out how to predict the relative size of a wildfire with scant data input.

1. Introduction

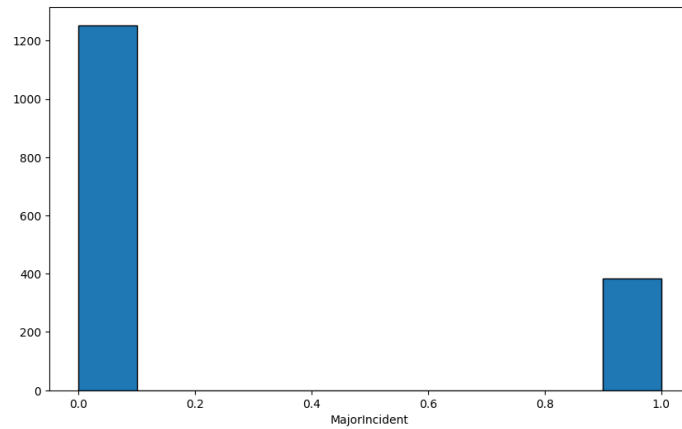
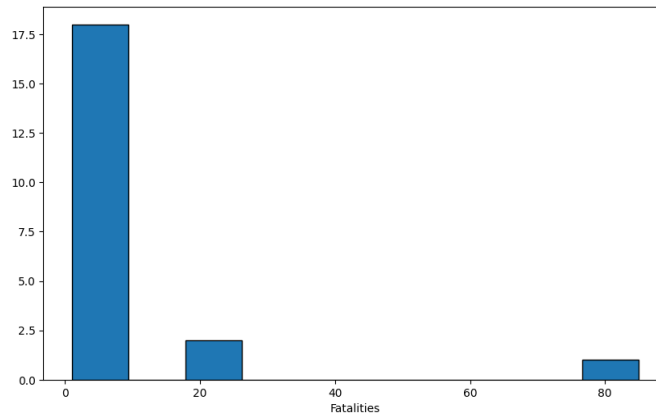
As climate change becomes an issue of global concern, arid, yet densely populated areas like California are prone to large-scale wildfires that cause immense amounts of economic and environmental damage. In 2023 alone, 7,127 fires in California caused a combined 324,917 acres of forest burned [2]. We attempt to alleviate this by predicting how much damage a certain wildfire in California will cause, in terms of acres burned, given chosen data points about the fire, in order to help with prevention, with the deployment of fire fighting means, and perhaps also with the safety of the personnel involved and of the population. To do this, we used a supervised regression model called XGBoost [3]. The dataset we used was the rather small numerical/categorical set of about 1,600 rows of information currently available on past wildfires in California and their characteristics like latitude, longitude, etc.

2. Background

At first, we considered analyzing a dataset that had been used to achieve a similar goal before, but decided against it, as the data was generalized to the entire United States. Its associated paper does not indicate whether the data set could be used to predict wildfire spread in other countries, or even only specific places in the United States. We wanted to have a more focused approach, and only predict the acres burned in California, as the United States is a very diverse and broad country, and it is more effective to specialize in a certain region with a more uniform climate.

3. Dataset

The data set that we used was a very small numerical set with 1,600 samples [4]. After tuning the amount of our data that we would use to test (an important figure given the small size of our set), we eventually found that it was optimal to use 10% as testing data and keep 90% for training data. In terms of data preprocessing, we had to cut many columns that frequently contained null values, and were left with only 'Latitude', 'Longitude', 'MajorIncident', 'PercentContained', 'CalFireIncident', 'Active', 'Started', 'Updated.' This left us with fewer columns to go off of but ultimately helped as it gave us the largest number of viable data points possible.



4. Methodology / Models

At first, we used a linear regression model, which gave us a very large mean absolute error of 8093. In an attempt to reduce this, we moved on to a random forest regression model, which gave us a better mean absolute error, and gave us a mean absolute error of 7855 after hyperparameter tuning. Finally, we searched for models that would give us a better result given a small dataset and decided to use XGBoost. XGBoost combines many simpler models called weak learners sequentially, where each of these focus on the inaccuracies of their predecessors, gradually improving the overall prediction accuracy of the model. XGBoost also incorporates regularization and tree pruning to prevent overfitting, ensuring the model generalizes well to new data [5]. Ultimately, using this model improved our results by a great amount, and after hyperparameter tuning, we arrived at a mean absolute error of around 3,350 acres burned.

5. Results and Discussion

The outcome of using an XGBoost model was a mean absolute error of around 3,350 acres burned. We got this figure through hyperparameter tuning, outlined by the spreadsheet below. To do this, we first took the default max_depth and N_estimators parameters and kept them as default while changing the learning rate, where we got our best error at a 0.2 learning rate. Next, we kept the learning rate the same at 0.2, and N_estimators at 100, and began changing the max_depth to the optimal depth of 7, which happened to be the default depth. Finally, we kept the learning_rate and max_depth parameters static while changing the N_estimators, finding the optimal value to be 125. This gave us a final pairing of 0.2, 7, and 130, leading us to our mean absolute error of about 3,350.

Learning_Rate	Max_Depth	N_Estimators	Result1	Result2		
0.1	7	100	3727	3797	DEFAULT	
0.2	7	100	3445	3410	BEST	
0.3	7	100	3973	4128		
0.05	7	100	3652	3679		
0.02	7	100	4374	4383		
0.075	7	100	3686	3721		
0.2	3	100	5621	5543		
0.2	4	100	4830	4813		
0.2	5	100	4367	4230		
0.2	6	100	3785	3889		
0.2	7	100	3445	3410	DEFAULT	BEST
0.2	8	100	3677	3677		
0.2	9	100	3810	3787		
0.2	10	100	3835	3765		
0.2	3	150	3472	3476		
0.2	4	100	3445	3410	DEFAULT	
0.2	5	125	3352	3360	BEST	
0.2	6	100	3427	3396		
0.2	7	130	3373	3471		

0.2	8	120	3485	3390		
Final Result:						
0.2	7	130	3373	3471	BEST	
Learning_Rate	Max_Depth	N_Estimators	Result1	Result2		
0.1	7	100	3727	3797	DEFAULT	
0.2	7	100	3445	3410	BEST	

6. Conclusions

To begin, we took a small dataset of 1,600 rows and began using linear regression in an attempt to predict the acres burned using other data points such as the latitude and longitude of the fire. This proved to have a very large mean absolute error of about 8,000 acres burned, so we used a random forest regression model, which decreased the error, but not by a significant margin. Finally, we used an XGBoost model, which after hyperparameter tuning, gave us our best mean absolute error of about 3,350. Additionally, further research could explore the incorporation of additional features such as wind speed, humidity, and vegetation density, which have been shown to influence wildfire behavior [6].

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References

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